

Exp No: 9 Identifying Suspicious Processes and Malware Analysis Using Sysinternals Process Explorer

DATE:

1. AIM

To conduct an in-depth behavioral and structural analysis of active operating system processes using Sysinternals Process Explorer; to evaluate hierarchical parent-child lineages, cryptographic digital signatures, and execution paths; to leverage cloud-based threat intelligence for automated multi-engine malware identification; and to successfully isolate, contain, and terminate unauthorized, masquerading, or anomalous processes to secure system integrity.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tool:** Sysinternals Process Explorer
- **Verification Utilities:** Web browser for online malware databases (VirusTotal, ProcessLibrary)

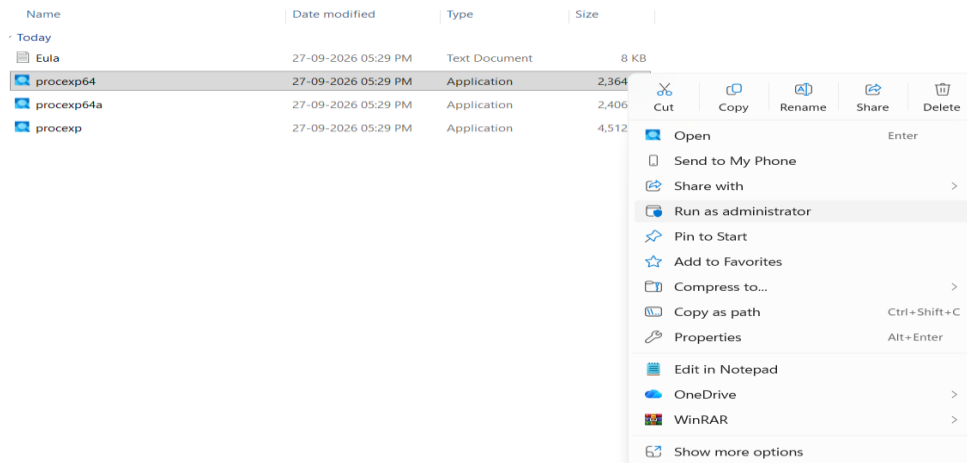
3. DESCRIPTION

Sysinternals Process Explorer is an advanced task manager replacement and live system monitoring tool for Windows that provides granular visibility into active processes, system resources, and security handles. It organizes applications into a hierarchical parent-child process tree and utilizes distinct color-coding to denote process states (e.g., pink for suspended, dark blue for system services, green for newly spawned instances). Beyond tracking dynamic resource consumption (CPU, memory, disk) and active TCP/IP connections, Process Explorer supports advanced threat-hunting features such as automated VirusTotal hash verification, digital signature validation, and lower-pane handle/DLL analysis. By inspecting image paths, security tokens, and execution lineage, security analysts can effectively distinguish legitimate system software from masquerading malware, rogue binaries, and unauthorized persistence mechanisms.

4. PROCEDURE

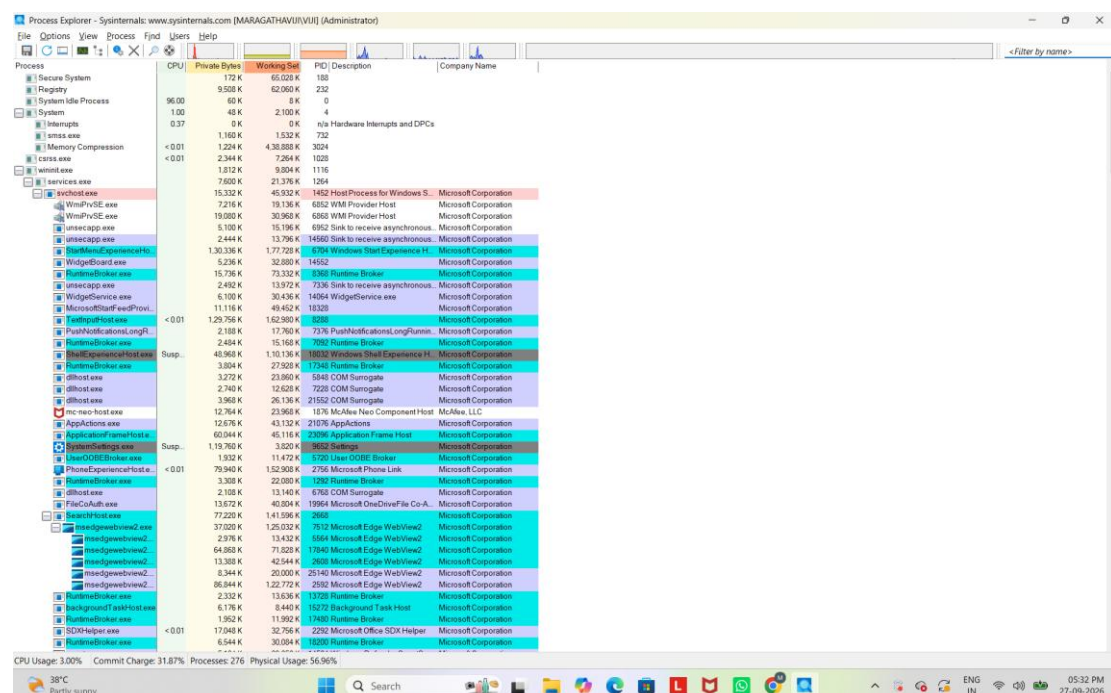
Step 1: Download and Initialization

1. Download the official Sysinternals Process Explorer suite from the Microsoft website and extract the archive to a local directory.
2. Navigate to the extracted folder, right-click procexp64.exe (for 64-bit systems), and select **Run as Administrator** to ensure full access to system-level process tokens and attributes.



Step 2: Interface Familiarization and State Analysis

1. Observe the main process tree interface displaying active processes, process IDs (PID), CPU/memory usage, descriptions, and company names.
2. Note the color-coding schema: Pink (suspended), Light Blue (user processes), Dark Blue (system services), and Green/Red (newly spawned or exited processes).



Step 3: Investigating Suspicious Processes

Perform targeted checks on unfamiliar or high-resource consumption processes:

- **Verify Digital Signatures:** Right-click the suspect process, select **Properties**, and check the **Image** tab for a valid digital signature from a trusted developer.
- **Check Execution Path:** Inspect the file path under the Image tab; legitimate system binaries must reside in system directories like C:\Windows\System32, whereas malware often runs from user profile temp folders or random directories.
- **Inspect Network Activity:** Navigate to the **TCP/IP** tab within the Properties window to identify unexpected or unauthorized external server connections.
- **Inspect Loaded DLLs (Lower Pane View):** Press Ctrl + D or go to View > Lower Pane View > DLLs. When you click a suspicious process in the upper tree, the lower pane will display all Dynamic Link Libraries loaded into that process. Look for unverified paths, missing company names, or DLL injection attempts where standard system processes load unexpected external libraries.

The screenshot displays the Process Explorer window from Sysinternals. The top pane shows a list of processes with columns for CPU, Private Bytes, Working Set, PID, Description, and Company Name. The bottom pane shows the loaded DLLs for the selected process, with columns for Name, Description, Company Name, and Path.

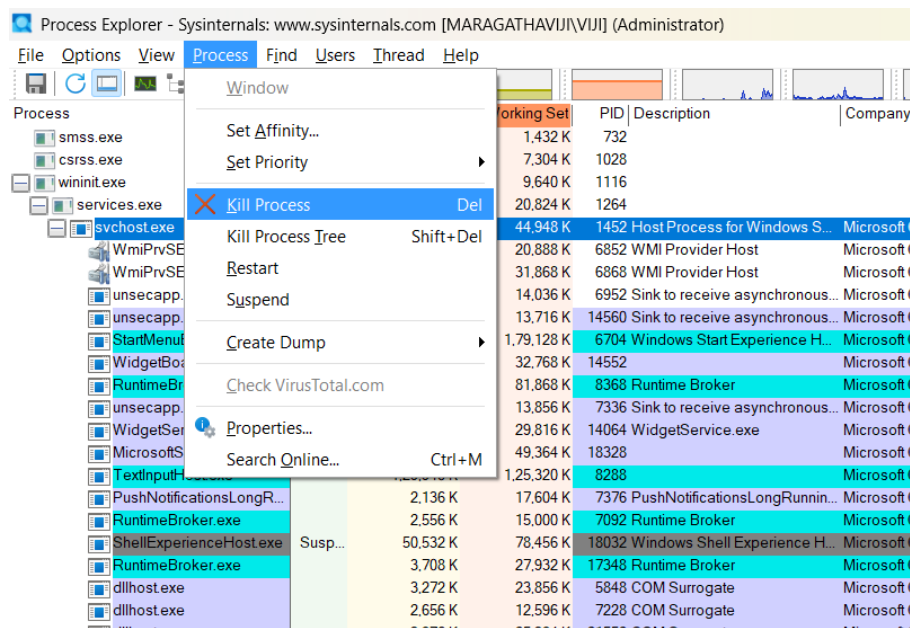
Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System	172 K	65,028 K	188			
Registry	9,528 K	61,096 K	232			
System Idle Process	97.90	60 K	0			
System	0.50	48 K	2,104 K	4		
smss.exe	1,160 K	1,516 K	732			
Memory Compression	1,232 K	4,88,052 K	3024			
csrss.exe	2,348 K	7,252 K	1028			
wininit.exe	1,812 K	9,804 K	1116			
services.exe	7,676 K	21,336 K	1264			
svchost.exe	15,244 K	45,716 K	1452		Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe	7,116 K	19,052 K	6852		WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe	18,380 K	38,580 K	6968		WMI Provider Host	Microsoft Corporation
unsecapp.exe	6,028 K	15,752 K	6952		Sink to receive asynchronous...	Microsoft Corporation
unsecapp.exe	2,440 K	13,800 K	14560		Sink to receive asynchronous...	Microsoft Corporation
StartMenuExperienceHost.exe	1,30,400 K	1,52,760 K	6704		Windows Start Experience H...	Microsoft Corporation
WidgetService.exe	5,280 K	32,396 K	14552			Microsoft Corporation
RuntimeBroker.exe	15,584 K	73,116 K	3560		Runtime Broker	Microsoft Corporation
unsecapp.exe	2,488 K	13,980 K	7336		Sink to receive asynchronous...	Microsoft Corporation
WidgetService.exe	6,060 K	30,152 K	14064		WidgetService.exe	Microsoft Corporation
MicrosoftStartFeedProv...	10,816 K	43,324 K	10328			Microsoft Corporation
PushNotificationsLongR...	129,794 K	1,51,076 K	3096			Microsoft Corporation
PushNotificationsLongR...	2,196 K	17,768 K	7376		PushNotificationsLongRunn...	Microsoft Corporation
RuntimeBroker.exe	2,484 K	15,160 K	7002		Runtime Broker	Microsoft Corporation

Name	Description	Company Name	Path
advapi32.dll	Advanced Windows 32 Base API	Microsoft Corporation	C:\Windows\System32\advapi32.dll
bcryptprimitives.dll	Windows Cryptographic Primitives	Microsoft Corporation	C:\Windows\System32\bcryptprimitives.dll
C:_1252.NLS			C:\Windows\System32\C_1252.NLS
C:_1252.NLS			C:\Windows\System32\C_1252.NLS
C:_1252.NLS			C:\Windows\System32\C_1252.NLS
C:_437.NLS			C:\Windows\System32\C_437.NLS
C:_437.NLS			C:\Windows\System32\C_437.NLS
C:_437.NLS			C:\Windows\System32\C_437.NLS
cbcatq.dll	COM+ Configuration Catalog	Microsoft Corporation	C:\Windows\System32\cbcatq.dll
code_explorer.com	Visual Studio Code	Microsoft Corporation	C:\Users\V\AppData\Local\Programs\Microsoft VS Code\11ba328a5a\code_explorer_command_x64.dll
combase.dll	Microsoft COM for Windows	Microsoft Corporation	C:\Windows\System32\combase.dll
dllhost.exe	COM Surrogate	Microsoft Corporation	C:\Windows\System32\dllhost.exe
gdi32.dll	GDI Client DLL	Microsoft Corporation	C:\Windows\System32\gdi32.dll
gdi32.dll	GDI Client DLL	Microsoft Corporation	C:\Windows\System32\gdi32.dll
imm32.dll	Multi-User Windows IMM32 API Clie...	Microsoft Corporation	C:\Windows\System32\imm32.dll
kernel.appcore.dll	AppModel API Host	Microsoft Corporation	C:\Windows\System32\kernel.appcore.dll
kernel32.dll	Windows NT BASE API Client DLL	Microsoft Corporation	C:\Windows\System32\kernel32.dll
KernelBase.dll	Windows NT BASE API Client DLL	Microsoft Corporation	C:\Windows\System32\KernelBase.dll
l_int.nls			C:\Windows\System32\l_int.nls
l_int.nls			C:\Windows\System32\l_int.nls
l_int.nls			C:\Windows\System32\l_int.nls
locale.nls			C:\Windows\System32\locale.nls
msvcp_win.dll	Microsoft® C Runtime Library	Microsoft Corporation	C:\Windows\System32\msvcp_win.dll
msvcp140.dll	Microsoft® C Runtime Library	Microsoft Corporation	C:\Windows\System32\msvcp140.dll
msvcrt.dll	Windows NT CRT DLL	Microsoft Corporation	C:\Windows\System32\msvcrt.dll

CPU Usage: 2.24% Commit Charge: 32.26% Processes: 270 Physical Usage: 55.81%

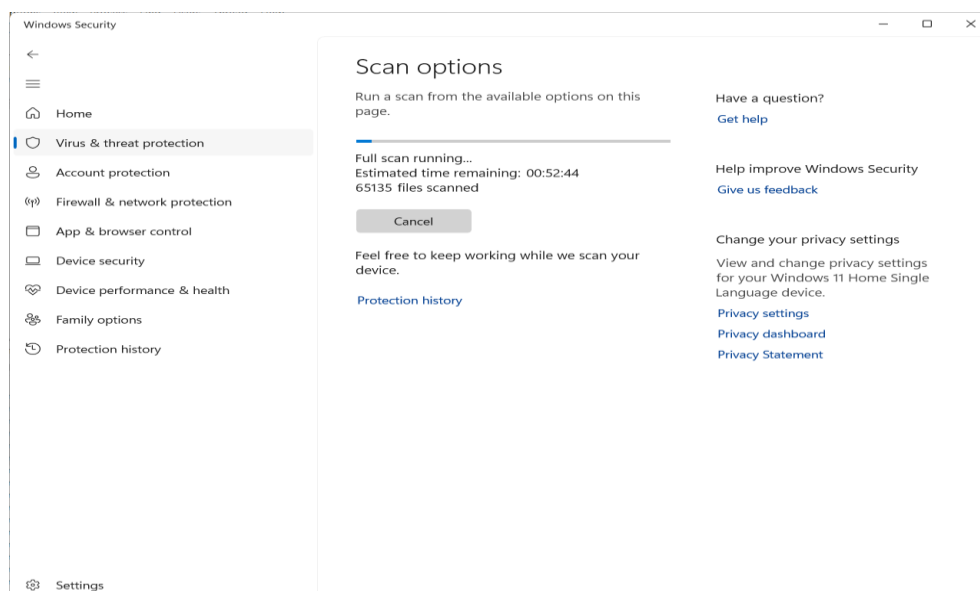
Step 5: Mitigation and Containment

1. If a process is confirmed malicious or unresponsive, right-click it and choose **Suspend** to halt its activity safely, or select **Kill Process** to terminate it immediately.
2. Locate the source executable using its verified path and remove the file.



Step 6: Comprehensive System Scan

1. Following threat mitigation, execute a full system scan using Windows Defender or dedicated anti-malware utilities (e.g., Malwarebytes) to ensure zero residual persistence.



5. OBSERVATIONS

- **Process Hierarchy & Lineage Integrity:** Legitimate Windows system processes (such as lsass.exe and winlogon.exe) correctly traced back to trusted system paths (C:\Windows\System32) and exhibited valid digital signatures from Microsoft Corporation. Parent-child relationships followed standard OS architecture (e.g., services.exe spawning child services).
- **Automated Threat Intelligence Integration:** Enabling VirusTotal integration allowed instant multi-engine cryptographic hash verification. Clean binaries yielded a 0/72 detection ratio, whereas anomalous binaries immediately flagged high detection ratios across multiple security vendors.
- **Network & Resource Correlation:** Anomalous processes flagged during investigation exhibited unauthorized outbound TCP/IP socket connections to external.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

6. RESULT

Sysinternals Process Explorer was successfully utilized to monitor active system processes, verify digital signatures and execution paths, and analyze process trees. Malicious or unauthorized processes were effectively identified, isolated, and terminated, ensuring system integrity and security.